

Auteur: Rob Willemsen
Studierichting: Informatica
Oefeningenreeks 4A nr. 7.4

$$\begin{aligned} & \int \frac{\sqrt[5]{x^3} + \sqrt[6]{x^5}}{\sqrt[4]{x}} dx \\ &= \int \frac{\sqrt[5]{x^3}}{\sqrt[4]{x}} dx + \int \frac{\sqrt[6]{x^5}}{\sqrt[4]{x}} dx \\ &= \int x^{\frac{3}{5} - \frac{1}{4}} dx + \int x^{\frac{5}{6} - \frac{1}{4}} dx \\ &= \int x^{\frac{7}{20}} dx + \int x^{\frac{7}{12}} dx \\ &= \frac{20}{27} x^{\frac{27}{20}} + \frac{12}{19} x^{\frac{19}{12}} + C \\ &= \frac{20}{27} \sqrt[20]{x^7} + \frac{12}{19} \sqrt[12]{x^9} + C \end{aligned}$$

References